Clay Emerson Wood, Ph.D.

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Research Profile

Experimental geoscientist and physicist who designs and builds high-pressure laboratory systems to measure how rock fractures, transmits fluids, and reacts under reservoir conditions. Produced experimental evidence of reaction-driven fracturing in ultramafic rock during CO_2 mineralization and established quantitative links between fracture structure and the hydraulic and elastodynamic properties of fractured rock — work spanning custom instrumentation, synchrotron X-ray CT, ultrasonic sensing, and HPC-scale analysis. Current research combines layered laboratory analogs and poroelastic modeling to investigate injection-induced deformation, motivated by wastewater disposal in the Delaware Basin. This work informs subsurface fluid injection, geothermal reservoir characterization, and thermal energy storage.

Education

Ph.D., Geosciences 2024
Pennsylvania State University, State College, PA
Dissertation — Imaging elastodynamic and hydraulic properties of fractured rock: Exploring effects of dynamic stressing, shearing, fracture aperture, and roughness

B.S., Physics 2016
Emory University, Atlanta, GA

Advisors

- **Current Postdoctoral advisor:** Prof. Hewei Tang (The University of Texas at Austin).
- **Postdoctoral advisors:** Prof. Nicola Tisato and Prof. Nicolás Espinoza (The University of Texas at Austin).
- **Ph.D. advisors:** Prof. Chris Marone (Pennsylvania State University & Dipartimento di Scienze della Terra, La Sapienza Università di Roma) and Prof. Parisa Shokouhi (Pennsylvania State University).

Relevant Technical Expertise

- **Experimental & instrumentation:** high-pressure flow-through and true-triaxial apparatus design and operation; custom hardware (CAD, 3D printing); Python and MATLAB instrument control and data acquisition; ultrasonic and X-ray CT integration.
- **Rock physics & flow:** permeability evolution, poromechanics, reactive transport, hydromechanical coupling under reservoir conditions.
- **Computation:** Python, Julia, MATLAB; signal/image processing, inverse and reduced-order modeling, workflow automation; HPC (TACC).
- **Characterization:** CT-based fracture analysis, elastic-wave/ultrasonic measurement, quantitative data interpretation.
- **Leadership:** multi-institutional research, external user facilities, graduate and undergraduate mentoring, peer-reviewed publications.

Professional Appointments

Postdoctoral Fellow – Experimental Poroelasticity 08/2026 – Present
The University of Texas at Austin, Austin, TX

- Establish laboratory capabilities and design layered gelatin–granular analog experiments to investigate injection-induced deformation, motivated by wastewater disposal in the Delaware Basin.
- Develop fluid-injection, optical imaging, and digital image correlation workflows to relate internal strain to surface displacement during controlled injection.
- Develop coupled poroelastic finite-element models in Julia to investigate how injection depth, layer thickness, permeability, and mechanical constraints govern pressure evolution and deformation.
- Mentor two undergraduate researchers in experimental poromechanics.

Geophysics Subject Matter Expert 03/2026 – Present
Handshake AI (Contract)

- Design evaluation questions, grading rubrics, and reference solutions used to benchmark and improve the scientific reasoning of frontier large language models in geosciences and physics.
- Evaluate model responses for technical accuracy in subsurface science, rock physics, and geophysics.

Postdoctoral Fellow – Reaction-Driven Cracking and Rock Characterization
The University of Texas at Austin, Austin, TX 09/2023 – 12/2025

- Designed and built high-pressure flow-through experiments on ultramafic rock (confining pressure up to 5 MPa, 100°C), resolving reaction-driven fracture propagation and permeability evolution during month-long reactive-flow experiments.
- Engineered integrated workflows coupling hydraulic systems, pressure/flow sensing, and X-ray CT to resolve transport and mechanical change in reacting rock.
- Built Python, Julia, and MATLAB pipelines to analyze terabytes of CT imaging data, interpret permeability measurements, and compare experiments.
- Co-advised one M.S. and one B.S. student on experimental rock mechanics research.

Graduate Researcher – Rock Mechanics Laboratory 07/2017 – 08/2023
Pennsylvania State University, State College, PA

- Designed and operated true-triaxial and flow-through experiments measuring permeability, elastic response, and hydromechanical coupling under controlled stress up to 20 MPa and pore pressure up to 5 MPa.
- Established quantitative relationships linking fracture aperture, roughness, and microstructure to hydraulic and geophysical properties, published across multiple *JGR: Solid Earth* papers.
- Ran synchrotron X-ray CT and ultrasonic experiments at the GeoSoilEnviroCARS beamline, Argonne National Laboratory, and analyzed fracture evolution and fluid transport.

Research Assistant – Fluid and Granular Mechanics Laboratory 06/2016 – 07/2017
Emory University, Atlanta, GA

- Ran precision experiments on fluid transport, wetting, and interfacial dynamics in multiphase systems, contributing to publications in *Physical Review Letters* and *Physical Review Fluids*.

Publications

Peer-reviewed articles and manuscripts under review or in preparation; first author on six listed works. Publications include *Physical Review Letters*, *Geophysical Research Letters*, and *JGR: Solid Earth*.

Google Scholar: 237 citations, h-index 8 (as of Nov. 2026)

C. Wood, A. Kenigsberg, J. Rivière, C. Marone, D. Saffer. (2026) Evolution of Permeability During Shear of Fault Gouge: Effects of Composition and Fabric. *In preparation.*

S. Almeshari, **C. Wood**, E. Ukar, A. Fall, N. Tisato. (2026) Carbon Mineralization in Ultramafic Rocks: Insights from Flow-Through CT-Imaged Experiments. *Geophys. Res. Lett., Under. Rev.*

C. Wood, N. Espinoza, N. Tisato, E. Ukar. (2026) Experimental evidence of reaction-driven fracturing and ultramafic rock weakening induced by CO_2 mineralization. *Geophys. Res. Lett.* doi:10.1029/2026GL124491.

E. Bozek, P. Borate, Chun-Yu Ke, **C. Wood**, L. Pillarisetti, M. Rivers, D. Elsworth, J. Rivière, P. Shokouhi. (2026) Linear and nonlinear elastodynamic signatures of fractured rock in relation to fault characteristics inferred from in-situ synchrotron X-ray computed tomography. *JGR: Solid Earth, Under review.* doi:10.26207/tqqc-4d73

C. Wood, J. Rivière, D. Elsworth, C. Marone, P. Shokouhi. (2026) Poromechanics of particle remobilization and interface stiffness of dynamically stressed fractured rock. *JGR: Solid Earth* doi:10.1029/2025JB033653

A. Eijsink, **C. Wood**, C. Marone, P. Shokouhi, J. Rivière, D. Elsworth. (2025) Evolution of frictional properties, permeability and elastic parameters due to surface roughness changes during fault maturation in laboratory shear experiments. *EPSL.* doi:10.1016/j.epsl.2025.119684

C. Wood, P. Manogharan, A. Rathbun, J. Rivière, D. Elsworth, C. Marone, P. Shokouhi. (2024) Relating fracture aperture to hydro-mechanical properties of dynamically-stressed tensile-fractured rock. *JGR: Solid Earth* doi:10.1029/2023JB027676

R. Affinito, **C. Wood**, S. Marty, D. Elsworth, C. Marone. (2024) The stability transition from stable to unstable frictional slip with finite pore pressure. *Geophys. Res. Lett.* doi:10.1029/2023GL105568

P. Manogharan, **C. Wood**, C. Marone, D. Elsworth, J. Rivière, P. Shokouhi. (2022) Experimental investigation of elastodynamic nonlinear response of dry intact, fractured and saturated rock. *Rock Mechanics & Rock Engineering* doi:10.1007/s00603-021-02552-6

S. Shreedharan, M. Ikari, **C. Wood**, D. Saffer, L. Wallace, C. Marone. (2022) Frictional and Lithological Controls on Shallow Slow Slip at the Northern Hikurangi Margin. *G-Cubed.* doi:10.1029/2021GC010107

C. Wood, P. Shokouhi, P. Manogharan, J. Rivière, D. Elsworth, C. Marone. (2021) Imaging Elastodynamic and Hydraulic Properties of In Situ Fractured Rock: An Experimental Investigation Exploring Effects of Dynamic Stressing and Shearing. *JGR: Solid Earth.* doi:10.1029/2020JB021521

P. Manogharan, **C. Wood**, C. Marone, D. Elsworth, J. Rivière, P. Shokouhi. (2021) Nonlinear elastodynamic behavior of intact and fractured rock under in-situ stress and saturation conditions. *JMPS.* doi:10.1016/j.jmps.2021.104491

P. Manogharan, **C. Wood**, C. Marone, D. Elsworth, J. Rivière, P. Shokouhi. (2021) Experimental Investigation of Elastodynamic Nonlinear Response of Dry Intact, Fractured and Saturated Rock. *Rock Mech Rock Eng.* doi:10.1007/s00603-021-02552-6

A. Mouat, **C. Wood**, J. Pye, J. Burton. (2020) Tuning Contact Line Dynamics and Deposition Patterns in Volatile Liquid Mixtures. *Phys. Rev. Lett.* doi:10.1103/PhysRevLett.124.064502

P. Shokouhi, J. Jin, **C. Wood**, J. Rivière, B. Madara, D. Elsworth, C. Marone. (2020). Dynamic stressing of naturally fractured rocks: On the relation between transient changes in permeability and elastic wave velocity. *Geophys. Res. Lett.* doi:10.1029/2019GL083557

A. Mouat, **C. Wood**, J. Pye, J. Burton. (2019) Liquid deposition through evaporation. *Phys. Rev. Fluids* doi:10.1103/PhysRevFluids.4.100512

J. Pye, **C. Wood**, J. Burton. (2018) Precursors to Molecular Slip on Smooth Hydrophobic Surfaces. *Phys. Rev. Lett.* doi:10.1103/PhysRevLett.121.134501

Selected Conference Talks & Presentations

C. Wood, N. Tisato, N. Espinoza, E. Ukar, A. Stokreef. Enhancing fracture permeability due to in-situ mineral carbonation: an experimental investigation into reaction-driven cracking in ultramafic rocks. Poster. American Geophysical Union Fall Meeting. December 2024.

C. Wood, C. Ke, J. Rivière, D. Elsworth, C. Marone, P. Shokouhi. Decoupling the poromechanics of particle remobilization and interface stiffness of dynamically stressed tensile fractured rock. Poster. American Geophysical Union Fall Meeting. December 2022.

C. Wood, C. Ke, A. Rathbun, J. Rivière, D. Elsworth, C. Marone, P. Shokouhi. Probing the micromechanical features of a fracture interface using a multi-physics approach: A numerical investigation relating asperity deformation with fluid flow. Talk. European Geosciences Union. May 2022.

C. Wood, P. Manogharan, J. Rivière, D. Elsworth, C. Marone, P. Shokouhi. Relating fracture aperture to hydraulic and elastodynamic properties of dynamically-stressed rock under true-triaxial stress conditions. Talk. American Geophysical Union Fall Meeting. New Orleans, LA, and online. December 2021.

C. Wood, P. Manogharan, J. Rivière, D. Elsworth, C. Marone, P. Shokouhi. Relating fracture aperture to hydraulic and elastodynamic properties of dynamically-stressed rock under true-triaxial stress conditions. Poster. The Physics of Earthquake Faulting. Rome, IT. September 2021.

C. Wood, P. Manogharan, S. Zi, J. Rivière, D. Elsworth, C. Marone, P. Shokouhi. The Relation Between Fracture Aperture and Hydro-mechanical Properties: An Experimental and Analytical Approach. Poster. American Geophysical Union Fall Meeting. Online. December 2020.

C. Wood, P. Manogharan, J. Jin, J. Rivière, D. Elsworth, C. Marone, P. Shokouhi. The Influence of Fracture Aperture on Permeability and Elastic Nonlinearity: An Integrated Analytical and Empirical Investigation. Talk. American Geophysical Union Fall Meeting. San Francisco, CA. December 2019.

C. Wood, B. Madara, P. Shokouhi, J. Jin, J. Rivière, D. Elsworth, C. Marone. The Effect of Roughness on the Elasticity and Permeability of Fractured Media. Poster. Schatzalp 3rd Induced Seismicity Workshop. Davos, CH. March 2019.

C. Wood, B. Madara, P. Shokouhi, J. Jin, J. Rivière, D. Elsworth, C. Marone. The Effect of Roughness on the Elasticity and Permeability of Fractured Media. Poster. American Geophysical Union Fall Meeting. Washington, D.C. December 2018.

C. Wood, B. Madara, P. Shokouhi, J. Jin, J. Rivière, D. Elsworth, C. Marone. The Effect of Roughness on the Elasticity and Permeability of Fractured Media. Poster. International School of Physics, Enrico Fermi. Varenna, IT. July 2018.

C. Wood, J. Pye, J. Burton. Liquid "Coffee Rings" and the Spreading of Volatile Liquid Mixtures. Talk. American Physical Society March Meeting. New Orleans, USA. March 2017.

C. Wood, J. Pye, J. Burton. Quantifying stick-slip contact line motion of evaporating sessile droplets. Talk. American Physical Society March Meeting. Baltimore, USA. March 2016.

Teaching

- GEOSC 597: Techniques in Geophysical Experimentation. Co-instructor with Dr. Chris Marone for a laboratory skills course.
- GEOSC 452: Hydrogeology. Teaching assistant. Instructors: Dr. Byron Parizek and Dr. John Hooker.

Professional Memberships

- American Geophysical Union member: 2018–Present.
- American Physical Society member: 2016–Present.

Honors & Awards

- Krynine Travel Grant, PSU Dept. of Geosciences 2018–2023
- Shell Geosciences Energy Research Facilitation Award, PSU Dept. of Geosciences 2019 & 2022
- Eagle Scout, Boy Scouts of America 2010
